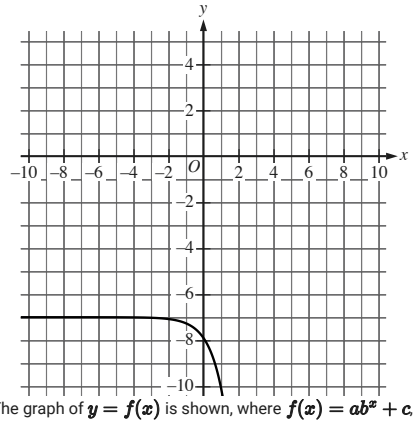
Question ID df71424b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: df71424b



The graph of $y = f(x)$ is shown, where $f(x) = ab^x + c$, and a, b, and c are constants. For how many values of x does $f(x) = 0$?

A. Three

B. Two

C. One

D. Zero

ID: df71424b Answer

Correct Answer:
D

Rationale

Choice D is correct. Each point (x, y) on the graph of $y = f(x)$ in the *xy*-plane gives a value of x and its corresponding value of $f(x)$, or y. For any value of x for which $f(x) = 0$, there is a corresponding point on the graph of $y = f(x)$ with a *y*-coordinate of 0. A point with a *y*-coordinate of 0 is a point where the graph intersects the *x*-axis. It's given that $f(x) = ab^x + c$, where a, b, and c are constants. In the *xy*-plane, the graph of an equation of this form will lie entirely either above or below the horizontal line defined by $y = c$. The part of the graph of $y = f(x)$ shown lies entirely below the horizontal line defined by $y = -7$, and thus the entire graph of $y = f(x)$ must lie below the line defined by $y = -7$. It follows that the graph of $y = f(x)$ will never intersect the *x*-axis. Therefore, there are zero values of *x* for which $f(x) = 0$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Question Difficulty:
Hard